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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

LOAN ELIGIBILITY VERIFICATION AND EMI PAYMENT TRACKING SYSTEM

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DECLARATION

I hereby declare that the project entitled **"LOAN ELIGIBILITY VERIFICATION AND EMI PAYMENT TRACKING SYSTEM"** submitted in partial fulfillment of the requirements for the course **Java Programming** is a **record of my original work** carried out under the guidance of **MUGILAN R**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____

(HARIPRIYA K)

CERTIFICATE

This is to certify that the project entitled **"LOAN ELIGIBILITY VERIFICATION AND EMI PAYMENT TRACKING SYSTEM"** is a bona fide work carried out by **HARIPRIYA K B.Sc. IT 'B' (2025-2028), INFORMATION TECHNOLOGY, KG College of Arts & Science,** in partial fulfillment of the requirements for the course **JAVA PROGRAMMING**

This project has been completed under my supervision and guidance during the academic period **April 2026.**

Place: COIMBATORE

Date:

Submitted for the Viva-Voce Examination held on _____

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ABSTRACT

The Loan Eligibility Verification and EMI Payment Tracking System is a comprehensive and efficient Java-based application developed to automate the process of loan approval and repayment monitoring. In today's fast-growing financial sector, banks and financial institutions handle a large number of loan applications daily. Managing these applications manually is time-consuming, error-prone, and inefficient.

This project aims to provide a digital solution that simplifies the loan approval process by automatically verifying the eligibility of applicants based on predefined criteria such as age, income, and loan amount. The system ensures that only eligible applicants proceed to the next stage, thereby reducing unnecessary workload and improving efficiency.

Once eligibility is confirmed, the system calculates the EMI (Equated Monthly Installment) using a standard financial formula. It provides users with detailed repayment information including monthly installment amount, total interest payable, and loan duration. Additionally, the system includes a payment tracking feature that allows users to monitor EMI payments and maintain financial discipline.

The application is developed using Core Java concepts such as object-oriented programming, classes, objects, loops, conditional statements, and exception handling. The system is designed to be user-friendly and reliable, ensuring smooth interaction and accurate results.

This project demonstrates the practical application of Java programming in financial systems and provides a foundation for developing advanced banking applications in the future

1.INTRODUCTION

1. The financial sector plays a crucial role in the economic development of a country. Loans are one of the most important financial services provided by banks and financial institutions. Individuals and businesses depend on loans for various purposes such as education, housing, business expansion, medical emergencies, and personal needs.
2. In traditional systems, loan processing involves multiple steps such as application submission, verification of documents, eligibility checking, EMI calculation, and repayment tracking. These processes are often carried out manually or with minimal automation, leading to inefficiencies and errors
3. To address these issues, the Loan Eligibility Verification and EMI Payment Tracking System has been developed. This system automates the process of loan eligibility verification and EMI calculation, ensuring accuracy and efficiency. It reduces the workload of employees and provides quick results to users.

The system allows users to enter their personal and financial details such as age, income, and loan amount. Based on these inputs, the system determines whether the user is eligible for a loan. If eligible, it calculates the EMI and provides repayment details. It also tracks EMI payments, helping users manage their finances effectively.

1.1 Objective

The primary objective of this project is to design and develop a robust, efficient, and user-friendly system that automates the process of loan eligibility verification and EMI payment tracking. The system aims to simplify complex financial processes and provide accurate results within a short time.

In modern financial systems, quick decision-making and accuracy are very important. This project focuses on achieving both by implementing a computerized solution that reduces manual effort and improves efficiency.

The detailed objectives of this project are as follows:

- To design and develop a system that can automatically verify loan eligibility based on user-provided details such as age, income, and loan amount.

- To implement an efficient EMI calculation mechanism using standard mathematical formulas to ensure accurate financial computations.
- To reduce dependency on manual calculations and minimize human errors in loan processing.
- To provide a simple, interactive, and user-friendly interface that allows users to easily interact with the system.
- To develop a payment tracking feature that helps users monitor their EMI payments and maintain proper financial discipline.
- To improve the overall efficiency and speed of loan processing systems.
- To demonstrate the practical application of Core Java concepts such as classes, objects, loops, conditional statements, and exception handling.
- To create a system that can be easily modified and enhanced in the future with additional features such as database connectivity and GUI.
- To provide a basic understanding of how financial applications are developed in real-world scenarios.

This project not only fulfills academic requirements but also provides practical knowledge that can be applied in real-world banking and financial systems.

1.2 Overview

The Loan Eligibility Verification and EMI Payment Tracking System is a comprehensive Java-based application designed to handle various loan-related operations in a simple and efficient manner. The system is developed as a menu-driven program, allowing users to interact with different functionalities step by step.

The application begins by collecting user information such as age, monthly income, loan amount, and loan duration. These inputs are then processed by the system to determine whether the user is eligible for a loan based on predefined conditions.

Once the eligibility is confirmed, the system proceeds to calculate the EMI using a standard financial formula. The calculated EMI represents the fixed monthly amount that the user needs to pay to repay the loan within the specified time period.

In addition to EMI calculation, the system also provides a payment tracking feature. This feature allows users to monitor their payment status, helping them keep track of paid and pending installments. This is particularly useful for maintaining financial discipline and avoiding missed payments.

The system is designed using a modular approach, where each functionality is divided into separate modules. This makes the application easy to understand, develop, and maintain. The modules work together to ensure smooth execution of the program.

Key highlights of the system include:

- Simple and easy-to-use interface
- Fast processing of user input
- Accurate calculation of EMI
- Efficient eligibility verification
- Structured and modular design

Overall, this system provides a basic yet effective solution for managing loan eligibility and EMI tracking. It serves as a foundation for developing more advanced financial management systems in the future

1.3 Features

The Loan Eligibility Verification and EMI Payment Tracking System provides wide range of features that enhance usability, accuracy, and efficiency. Each feature is designed to simplify loan processing and provide a better user experience.

1. Automated Loan Eligibility Verification

The system automatically evaluates whether a user qualifies for a loan based on predefined rules such as age, income, and loan amount. This remove manual verification and ensures:

Faster decision-making

Consistent results

Reduced chances of human error

Example: A user with income below the required threshold is immediately marked as not eligible.

2. Accurate EMI Calculation

The system uses a standard financial formula to calculate EMI high precision. It considers:

- Principal amount
- Interest rate
- Loan duration

Benefits:

- Eliminates manual calculation errors
- Provides exact monthly installment
- Helps users plan finances effectively

3. Detailed Financial Output

In addition to EMI, the system provides:

- Total amount payable
- Total interest amount
- Monthly payment details

This helps users clearly understand their financial commitment before taking a loan.

4. User-Friendly Interface

The system is designed as a menu-driven console application with:

- Clear instructions
- Step-by-step interaction
- Simple navigation
- Even users with basic computer knowledge can easily operate the system.

2. PROGRAM ANALYSIS

Program analysis is a crucial phase in software development where the existing system is studied in detail and the requirements of the new system are identified. This phase helps in understanding the problem clearly and provides a strong foundation for designing and developing an efficient solution.

In this project, program analysis focuses on identifying the issues in traditional loan processing systems and proposing an automated solution that improves accuracy, efficiency, and user experience.

2.1 Problem Definition

The traditional loan processing system is mostly manual and involves several steps such as collecting applicant information, verifying eligibility, calculating EMI, and tracking repayments. These tasks are usually performed by bank employees using paper records or basic tools like calculators and spreadsheets.

This manual approach leads to several challenges and limitations:

- High chances of calculation errors: Manual EMI calculations may result in incorrect values due to human mistakes.
- Time-consuming process: Loan approval takes longer as each step is handled manually.
- Difficulty in handling multiple applications: Managing a large number of loan applications becomes complex and inefficient.
- Lack of proper record management: Maintaining records manually increases the risk of data loss and mismanagement.
- Poor customer experience: Delays and errors in loan processing can lead to dissatisfaction among customers.

For example, if a bank employee incorrectly calculates the EMI or approves a loan for an ineligible customer, it can lead to financial losses and credibility issues.

Therefore, there is a strong need for an automated system that can perform these operations quickly, accurately, and efficiently.

2.2 Project Overview

The Loan Eligibility Verification and EMI Payment Tracking System is designed to overcome the limitations of manual loan processing systems.

The system allows users to enter their personal and financial details such as age, income, loan amount, and duration. Based on these inputs, the system verifies whether the user meets the eligibility criteria.

The eligibility criteria may include:

- Minimum monthly income requirement
- Age limits (e.g., 21 to 60 years)
- Maximum loan amount restrictions

If the user satisfies all conditions, the system approves the loan and proceeds to calculate the EMI using a standard formula. The calculated EMI represents the amount the user must pay every month.

The system also includes a payment tracking feature that allows users to monitor their EMI payments. This helps users keep track of their financial commitments and avoid missing payments.

The application is developed using Java programming language and follows object-oriented principles. It is designed as a menu-driven system, making it easy to use even for non-technical users.

Overall, the system provides a simple, efficient, and reliable solution for managing loan eligibility and EMI tracking.

2.3 Scope and Limitations

Scope

The scope of the Loan Eligibility Verification and EMI Payment Tracking System includes various functionalities that can be used in real-world scenarios. The system can be applied in:

Banks and financial institutions: To automate loan processing and reduce manual workload.

Educational purposes: To demonstrate the application of Java programming in financial systems.

Small financial organizations: To manage loan eligibility and EMI calculations efficiently.

Personal use: Individuals can use the system to estimate their loan eligibility and EMI before applying for a loan.

The system provides the following capabilities:

- Instant loan eligibility verification
- Accurate EMI calculation
- Simple user interaction
- Basic payment tracking functionality

This makes the system useful as a foundational tool for financial management applications.

Limitations

Despite its advantages, the system has certain limitations:

- No database integration: The system does not store user data permanently.
- Limited validation rules: Only basic eligibility conditions are considered.
- No graphical user interface (GUI): The system is console-based.
- No real-time banking integration: It does not connect with actual banking systems.
- Basic payment tracking: Advanced features like reminders and notifications are not included.

These limitations can be addressed in future enhancements by adding advanced features such as database connectivity, GUI, and online integration.

3.SYSTEM DESIGN

System design is a crucial stage in the development of any software application. It involves defining the architecture, components, modules, interfaces, and data flow of the system. A well-planned system design ensures that the application is efficient, reliable, scalable, and easy to maintain.

In the Loan Eligibility Verification and EMI Payment Tracking System, the design is created using a structured and modular approach. The system is divided into smaller units called modules, where each module performs a

specific function. This makes the system organized, reduces complexity, and improves overall performance.

The system is implemented as a menu-driven application that allows users to interact step-by-step. Each operation is clearly defined, making the system simple and easy to use even for beginners

3.1 Modules

1. User Input Module

This module acts as the interface between the user and the system. It collects all required details from the user.

Details Collected:

- Name
- Age
- Monthly income
- Loan amount
- Loan duration

Responsibilities:

- Accept input from user
- Validate input values
- Ensure data accuracy

Importance: This module ensures that the system receives correct and meaningful data, which is essential for accurate processing.

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3. EMI Calculation Module

This module calculates the monthly installment using a standard financial formula.

Formula: EMI = [P × R × (1+R)^N] / [(1+R)^N – 1]

Inputs:

- Principal (loan amount)
- Interest rate
- Time period

Outputs:

- Monthly EMI
- Total amount payable
- Total interest

Additional Explanation: This module performs complex mathematical calculations using Java functions. It ensures precision and eliminates errors.

4. Payment Tracking Module

This module helps users track their loan repayment progress.

Functions:

- Records number of EMIs paid
- Displays remaining EMIs

- Shows payment status

Benefits:

- Helps users stay organized
- Avoids missed payments

Future Improvements:

- Payment history storage
- Alerts and reminders

5. Exit Module

This module allows the user to safely terminate the application.

Functions:

- Ends program execution
- Ensures proper closure

3.2 System Architecture

The system follows a simple layered architecture:

1. Input Layer – Accepts user data
2. Processing Layer – Performs calculations and decisions
3. Output Layer – Displays results

This structure ensures clear separation of responsibilities.

3.3 Data Flow Description

The flow of data in the system follows a logical sequence:

1. User provides input
2. Input is validated
3. Eligibility is checked
4. EMI is calculated
5. Results are displayed
6. Payment tracking updates status

3.4 UML Design (Conceptual)

The system can be represented using basic UML concepts:

Classes:

- User
- Loan
- EMI Calculator

Interactions:

- User inputs data
- Loan processes eligibility
- EMI Calculator computes installments

3.5 Interface Design

The system uses a simple console interface:

- Menu-driven options
- Clear instructions
- Easy navigation

Example Menu:

1. Enter Details
2. Check Eligibility
3. Calculate EMI
4. Track Payment
5. Exit

3.6 Error Handling Design

The system includes mechanisms to handle errors effectively:

- Invalid input detection
- Error messages display

- Prevention of system crash

3.7 Assumptions

- User provides correct input
- Interest rate remains constant
- No external system integration

3.8 Advantages of System Design

- Modular structure
- Easy maintenance
- Better performance
- Improved reliability
- Scalable system

3.9 Limitations of Design

- No GUI support
- No database storage
- Limited validation rules

3.10 Overall Workflow

1. Start system
2. Display menu
3. Get user input
4. Validate data
5. Check eligibility
6. Calculate EMI
7. Display output
8. Track payment
9. Exit system

4.METHODOLOGY

Methodology refers to the systematic, structured, and well-organized approach followed to design and develop the Loan Eligibility Verification and EMI Payment Tracking System. It plays a vital role in ensuring that the project is completed in a logical sequence and meets all the functional and technical requirements.

A proper methodology helps in breaking down the entire development process into smaller manageable phases. Each phase contributes to the successful completion of the project and ensures that the final system is accurate, efficient, and user-friendly.

The methodology used in this project follows a step-by-step development life cycle, which includes problem analysis, system design, program development, testing and validation, and result documentation. Each stage is explained in detail below.

4.1 Problem Analysis

Problem analysis is the initial and most critical stage of the project development process. In this phase, a detailed study of the existing loan processing system was carried out to identify its weaknesses and limitations.

Traditional loan systems rely heavily on manual operations such as verifying applicant data, calculating EMI manually, and maintaining repayment records on paper or spreadsheets. These processes are not only time-consuming but also prone to human errors.

Some of the major problems identified are:

Manual calculations leading to incorrect EMI values

Delay in loan approval due to time-consuming verification

Difficulty in handling multiple loan applications simultaneously

Lack of proper tracking system for EMI payments

Increased workload for employees

For example, if an employee miscalculates the interest rate or loan duration, it can lead to incorrect EMI values, causing financial issues for both the bank and the customer.

After identifying these problems, the requirements for the new system were defined. **The** goal was to develop a system that is:

- Accurate in calculations
- Fast in processing
- Easy to use
- Reliable and efficient

This phase helped in clearly understanding what the system should achieve and what features must be included.

4.2 System Design

System design is the phase where the overall structure of the application is planned before coding begins. A well-designed system ensures smooth implementation and reduces complexity during development.

In this project, a modular design approach was used. The entire system was divided into smaller modules, each responsible for a specific function. This approach improves readability, maintainability, and scalability.

The major modules designed are:

- User Input Module
- Eligibility Verification Module
- EMI Calculation Module
- Payment Tracking Module
- Exit Module

Each module was carefully planned with defined inputs, processing logic, and outputs. For example:

The User Input Module collects and validates user data

The Eligibility Module checks conditions like income and age

The EMI Module performs mathematical calculations

The system flow was also designed to ensure smooth navigation. The user starts from the main menu, enters details, checks eligibility, and then proceeds to EMI calculation.

Additionally, design considerations such as simplicity, efficiency, and user-friendliness were taken into account. The interface was kept simple so that even non-technical users can operate the system easily.

4.3 Program Development

Program development is the phase where the system is implemented using programming language. In this project, Java was used due to its object-oriented features, platform independence, and robustness.

The development process involved several steps:

- Creating classes to represent system modules
- Writing methods for specific functionalities
- Implementing decision-making using if-else statements
- Using loops for repeated operations
- Handling user input using Scanner class

For example, eligibility checking was implemented using conditional statements such as:

if (income > 25000 && age > 21 && age < 60)

This ensures that only eligible users can proceed to the next stage.

The EMI calculation was implemented using the standard formula. The program takes inputs such as loan amount, interest rate, and time period, and calculates EMI accurately.

Exception handling was also included to manage runtime errors such as invalid input. This ensures that the program does not crash and provides meaningful error messages to the user.

The code was written in a structured and modular format, making it easy to understand and maintain. Proper indentation and comments were used to improve readability.

4.4. Testing and Validation

Testing is an essential phase that ensures the correctness and reliability of the system. In this phase, the developed application was tested using various inputs to verify its functionality.

Example test cases:

Test Case	Input	Expected Output
Valid Data	**Income = 30000**	**Eligible**
Invalid Data	**Age = 15**	**Not Eligible**
EMI Check	**Loan = 100000**	**Correct EMI**

During testing, errors such as incorrect calculations or improper input handling were identified and corrected. This process improved the accuracy and performance of the system.

Testing also ensured that the system is user-friendly and can handle real-world scenarios effectively

Different types of testing performed include:

1. **Functional Testing –** Ensures that each module performs its intended function correctly.
2. **Input Testing –** Checks system behavior with valid and invalid inputs.
3. **Boundary Testing** – Tests extreme values such as minimum and maximum limits.
4. **Error Handling Testing –** Ensures proper handling of incorrect inputs.

4.5 Result and Documentation

After successful testing, the results of the system were analyzed and documented. The system performed all required functions such as eligibility verification and EMI calculation accurately.

The results showed that:

The system provides quick and accurate results

It reduces manual effort significantly

It improves efficiency in loan processing

Documentation plays an important role in project development. It includes detailed information about system design, implementation**,** and testing.

Screenshots of program execution were included to demonstrate the working of the system. Proper documentation helps future developers understand the system easily and make improvements if required.

4.6. Algorithm

The algorithm represents the logical steps followed by the system to perform its operations. It provides a clear understanding of how the program works internally.

Detailed Algorithm:

1. Start the program
2. Display menu options to the user
3. Ask the user to enter details (age, income, loan amount, duration)
4. Validate the input values
5. Check eligibility conditions:
    - If income >= required value AND age within limit
    - Then user is eligible
    - Else user is not eligible
6. If eligible:
    - Calculate EMI using formula
    - Display EMI amount

- Show repayment details

7. Provide option for payment tracking
8. Display payment status (paid/pending)
9. End the program

This algorithm ensures a clear flow of execution and helps in writing the program efficiently

4.7. Flowchart

The flowchart is a graphical representation of the system process. It visually shows the sequence of steps involved in the program.

Flowchart Description:

Start
↓
Display Menu
↓
Enter User Details
↓
Validate Input
↓
Check Eligibility
↓
Is Eligible?
→ No → Display Not Eligible → End
→ Yes → Calculate EMI
↓
Display EMI Details
↓
Track Payment Status
↓
End

The flowchart helps in understanding the program logic easily and is useful for both developers and users.

5.SOURCE CODE

```
import java.util.Scanner;

public class LoanSystem {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String name;
        int age, duration;
        double income, loan, emi;
        double rate = 10.0 / 12 / 100; // monthly interest

        // Input
        System.out.println("Enter Name:");
        name = sc.nextLine();

        System.out.println("Enter Age:");
        age = sc.nextInt();

        System.out.println("Enter Monthly Income:");
        income = sc.nextDouble();

        System.out.println("Enter Loan Amount:");
        loan = sc.nextDouble();

        System.out.println("Enter Loan Duration (months):");
        duration = sc.nextInt();

        // Eligibility Check
        if (income >= 25000 && age >= 21 && age <= 60) {

            System.out.println("Eligible for Loan");

            // EMI Calculation
            emi = (loan * rate * Math.pow(1 + rate, duration)) /
                  (Math.pow(1 + rate, duration) - 1);

            System.out.println("Monthly EMI: " + emi);
```

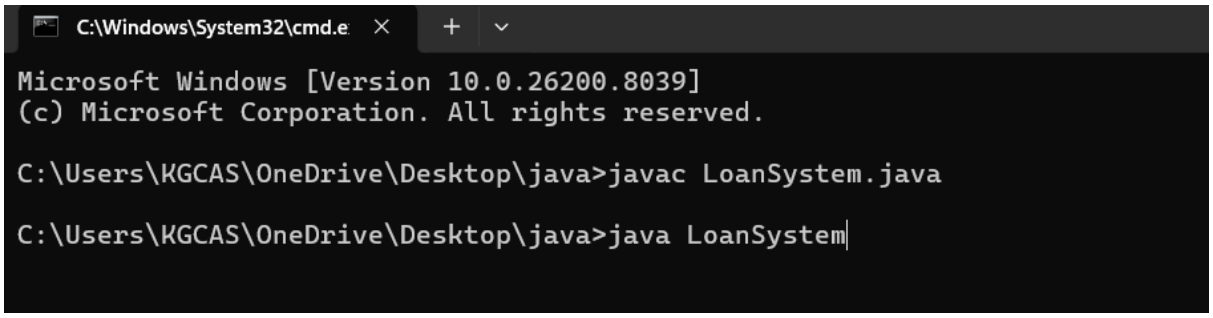
```
            // Payment Tracking
            System.out.println("Enter number of EMIs paid:");
            int paid = sc.nextInt();

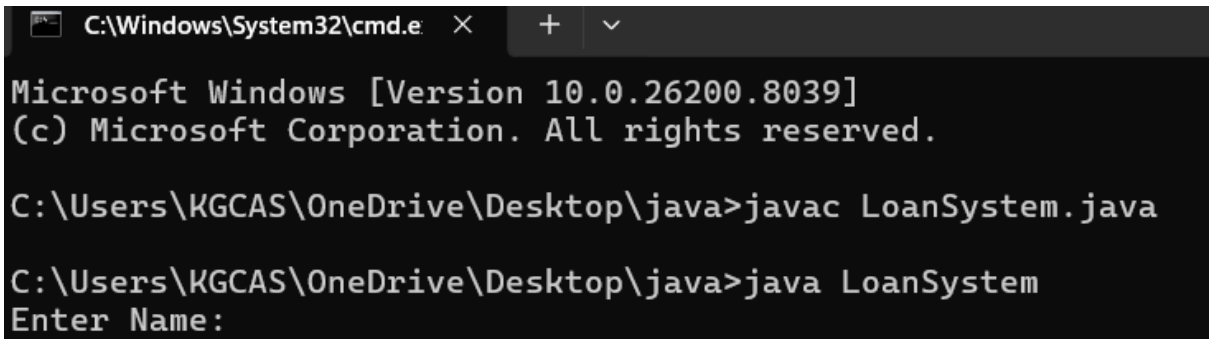
            System.out.println("Remaining EMIs: " + (duration - paid));

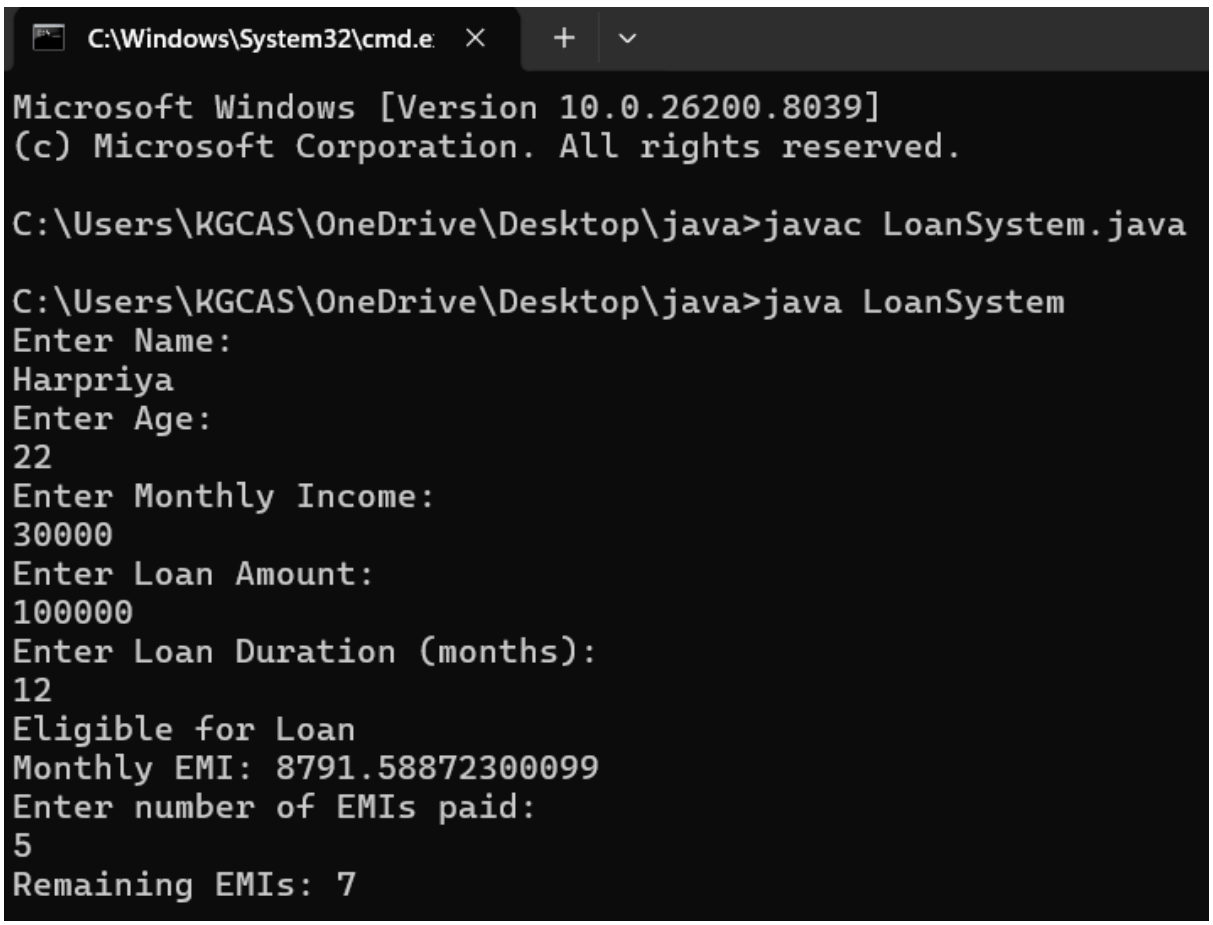
        } else {
            System.out.println("Not Eligible for Loan");
        }

        sc.close();
    }
}
```

6.INPUT AND OUTPUT SCREENS







7.DISCUSSION & CONCLUSION

7.1 Discussion

The Loan Eligibility Verification and EMI Payment Tracking System was developed to automate loan processing tasks such as eligibility checking and EMI calculation. The system was tested with various inputs and was found to produce accurate and reliable results.

The modular design helped in organizing the system efficiently, and each module performed its function correctly. The application reduces manual effort and improves the speed of loan-related operations. The simple interface makes it easy for users to interact with the system.

Overall, the system proves that basic financial operations can be handled effectively using Java, providing faster results compared to manual methods.

7.2 Conclusion

The project successfully demonstrates how Java can be used to develop a basic financial application. It reduces errors, saves time, and improves efficiency in loan processing.

The system can be further improved by adding features such as a graphical user interface, database connectivity, and advanced payment tracking. These enhancements will make the system more suitable for real-world applications.

In conclusion, this project provides a strong foundation for developing more advanced financial systems and enhances practical programming skills.

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The system can be further improved by adding features such as a graphical user interface, database connectivity, and advanced payment tracking.

8.REFERENCES

The following resources were used during the development of this project:

Books

- **Herbert Schildt – "Java: The Complete Reference"**
- **Y. Daniel Liang – "Introduction to Java Programming"**

- ## Websites

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